Agentic AI Workflow

1. Beginner Level Workflow

Goal

Understand how AI Agents work with prompts, tools, and APIs.

Workflow

User Input
↓
LLM (GPT/Claude/Llama)
↓
Prompt Processing
↓
Tool/API Call
↓
Response Generation
↓
User Output

Technologies

- Python
- OpenAI API
- FastAPI
- Streamlit
- REST APIs
- JSON

Beginner Projects

1. **FAQ Chatbot**
2. Weather Agent
3. Currency Converter Agent
4. News Search Agent
5. Resume Screening Agent
6. Customer Support Agent
7. PDF Question Answering Agent
8. Email Summarization Agent

2. Intermediate Level Workflow

Goal

Build agents with memory, RAG, and multiple tools.

Workflow

User Query
↓
Agent
↓
Task Planning
↓
Memory Check
↓
Vector Database Search
↓
Tool Selection
↓
API Execution
↓
Result Evaluation
↓
Final Response

Technologies

- LangChain
- LangGraph
- CrewAI
- ChromaDB
- FAISS
- Pinecone
- FastAPI
- PostgreSQL
- Docker

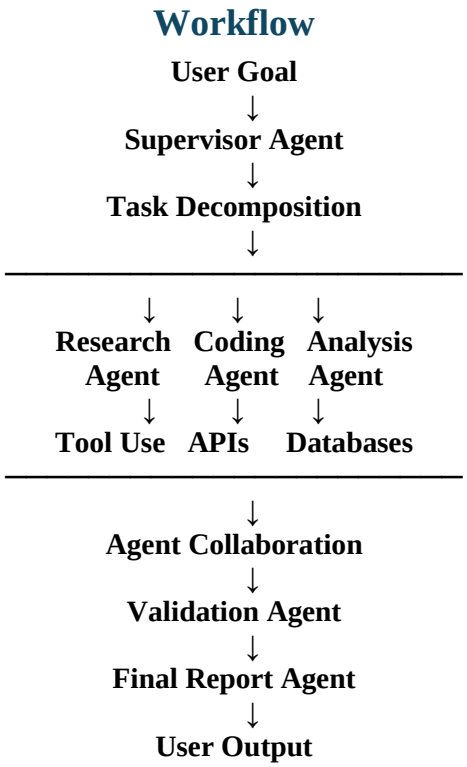
Intermediate Projects

1. Travel Planning Agent
2. Financial Research Agent
3. HR Recruitment Agent
4. Legal Document Assistant
5. Medical Information Assistant
6. Customer Feedback Analysis Agent
7. Personal Productivity Agent
8. AI Coding Assistant

3. Advanced Level Workflow

Goal

Create autonomous multi-agent systems.



Technologies

- CrewAI
- LangGraph
- AutoGen
- OpenAI Agents SDK
- MCP (Model Context Protocol)
- Vector Databases
- Kubernetes
- Docker
- Redis
- RabbitMQ

Advanced Projects

1. Multi-Agent HR System
2. Autonomous Research Agent

3. Stock Market Analysis Agent
4. AI Software Development Team
5. AI Business Consultant
6. AI Cybersecurity Analyst
7. AI Data Science Team
8. AI Content Creation Team

4. Enterprise-Level Agentic AI Workflow

User Goal
↓
Orchestrator Agent
↓
Planner Agent
↓
Task Breakdown
↓

Research Agent
Coding Agent
Database Agent
Analytics Agent
Security Agent
Documentation Agent

↓
Knowledge Base (RAG)
↓
Vector Database
↓
Memory Layer
↓
Quality Check Agent
↓
Human Approval
↓
Deployment Agent
↓
Final Output

Enterprise Tools

- OpenAI Agents SDK
- CrewAI
- LangGraph
- AutoGen
- ChromaDB

- Pinecone
- Weaviate
- PostgreSQL
- Redis
- Docker
- Kubernetes
- AWS
- Azure
- GCP

Learning Path

Python
↓
APIs & JSON
↓
Prompt Engineering
↓
LLMs
↓
RAG
↓
Vector Databases
↓
LangChain
↓
AI Agents
↓
CrewAI
↓
LangGraph
↓
Multi-Agent Systems
↓
MCP
↓
Agentic AI
↓
Enterprise Agentic AI

This progression takes you from a simple single-agent chatbot to a fully autonomous multi-agent Agentic AI system.